

Labsheet- 2

Exploratory Data Analysis(EDA)

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("Sample - Superstore.csv", encoding="latin1")

# Question 1: EDA - Explore dataset structure
print("Dataset Shape:")
print(df.shape)

print("\nDataset Information:")
df.info()

print("\nStatistical Summary:")
print(df.describe())

# Question 2: Univariate, Bivariate and Multivariate Analysis

# Univariate Analysis
plt.figure(figsize=(8, 5))
sns.histplot(df["Sales"], kde=True)
plt.title("Univariate Analysis - Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

# Bivariate Analysis
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="Sales", y="Profit")
plt.title("Bivariate Analysis - Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()

# Multivariate Analysis
numerical_columns = df.select_dtypes(include=np.number).columns
print("\nNumerical Columns:")
print(numerical_columns.tolist())

# Question 3: Correlation Heatmap

correlation = df[numerical_columns].corr()

print("\nCorrelation Matrix:")
print(correlation)

plt.figure(figsize=(10, 6))
sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```

```
# Question 4: Histograms, Box Plots and Scatter Plots

# Histogram
plt.figure(figsize=(8, 5))
sns.histplot(df["Profit"], kde=True)
plt.title("Histogram of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()

# Box Plot
plt.figure(figsize=(8, 5))
sns.boxplot(y=df["Profit"])
plt.title("Box Plot of Profit")
plt.ylabel("Profit")
plt.show()

# Scatter Plot
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="Sales", y="Profit")
plt.title("Scatter Plot - Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()

# Question 5: Identify Data Quality Issues

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Records:")
print(df.duplicated().sum())

print("\nData Types:")
print(df.dtypes)

print("\nUnique Values:")
print(df.nunique())

# Question 6: Correlation for Predictive Analytics

print("\nCorrelation with Profit:")
print(df[numerical_columns].corr()["Profit"].sort_values(ascending=False))

plt.figure(figsize=(8, 5))
sns.heatmap(
    df[numerical_columns].corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation for Predictive Analysis")
plt.show()

# Question 7: Visualization of Categorical and Numerical Variables
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```
# Categorical Variable
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x="Category")
plt.title("Categorical Variable - Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()

# Numerical Variable
plt.figure(figsize=(8, 5))
sns.histplot(df["Sales"], kde=True)
plt.title("Numerical Variable - Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

# Question 8: Business Insights from Superstore Dataset

category_sales = df.groupby("Category")["Sales"].sum()
category_profit = df.groupby("Category")["Profit"].sum()

region_sales = df.groupby("Region")["Sales"].sum()
region_profit = df.groupby("Region")["Profit"].sum()

segment_sales = df.groupby("Segment")["Sales"].sum()
segment_profit = df.groupby("Segment")["Profit"].sum()

print("\nSales by Category:")
print(category_sales)

print("\nProfit by Category:")
print(category_profit)

print("\nSales by Region:")
print(region_sales)

print("\nProfit by Region:")
print(region_profit)

print("\nSales by Segment:")
print(segment_sales)

print("\nProfit by Segment:")
print(segment_profit)

print("\nHighest Sales Category:")
print(category_sales.idxmax())

print("\nHighest Profit Category:")
print(category_profit.idxmax())

print("\nHighest Sales Region:")
print(region_sales.idxmax())

print("\nHighest Profit Region:")
print(region_profit.idxmax())
```

```
# Question 9: Feature Selection and Model Building

# Correlation of numerical features with Profit
feature_correlation = df[numerical_columns].corr()["Profit"].abs().sort_values(a

print("\nFeature Correlation with Profit:")
print(feature_correlation)

# Highly correlated features
print("\nFeatures with correlation greater than 0.3:")
print(feature_correlation[feature_correlation > 0.3])

# Question 10: Handling Large-Scale Dataset Challenges

# Check dataset size
print("\nNumber of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

# Check memory usage
print("\nMemory Usage:")
print(df.memory_usage(deep=True).sum() / (1024 ** 2), "MB")

# Analyze a sample of large dataset
sample_df = df.sample(
    min(1000, len(df)),
    random_state=42
)

print("\nSample Dataset Shape:")
print(sample_df.shape)

print("\nEDA completed successfully!")
```

Dataset Shape:
(9994, 21)

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 9994 entries, 0 to 9993

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	Row ID	9994 non-null	int64
1	Order ID	9994 non-null	object
2	Order Date	9994 non-null	object
3	Ship Date	9994 non-null	object
4	Ship Mode	9994 non-null	object
5	Customer ID	9994 non-null	object
6	Customer Name	9994 non-null	object
7	Segment	9994 non-null	object
8	Country	9994 non-null	object
9	City	9994 non-null	object
10	State	9994 non-null	object
11	Postal Code	9994 non-null	int64
12	Region	9994 non-null	object
13	Product ID	9994 non-null	object
14	Category	9994 non-null	object
15	Sub-Category	9994 non-null	object
16	Product Name	9994 non-null	object
17	Sales	9994 non-null	float64
18	Quantity	9994 non-null	int64
19	Discount	9994 non-null	float64
20	Profit	9994 non-null	float64

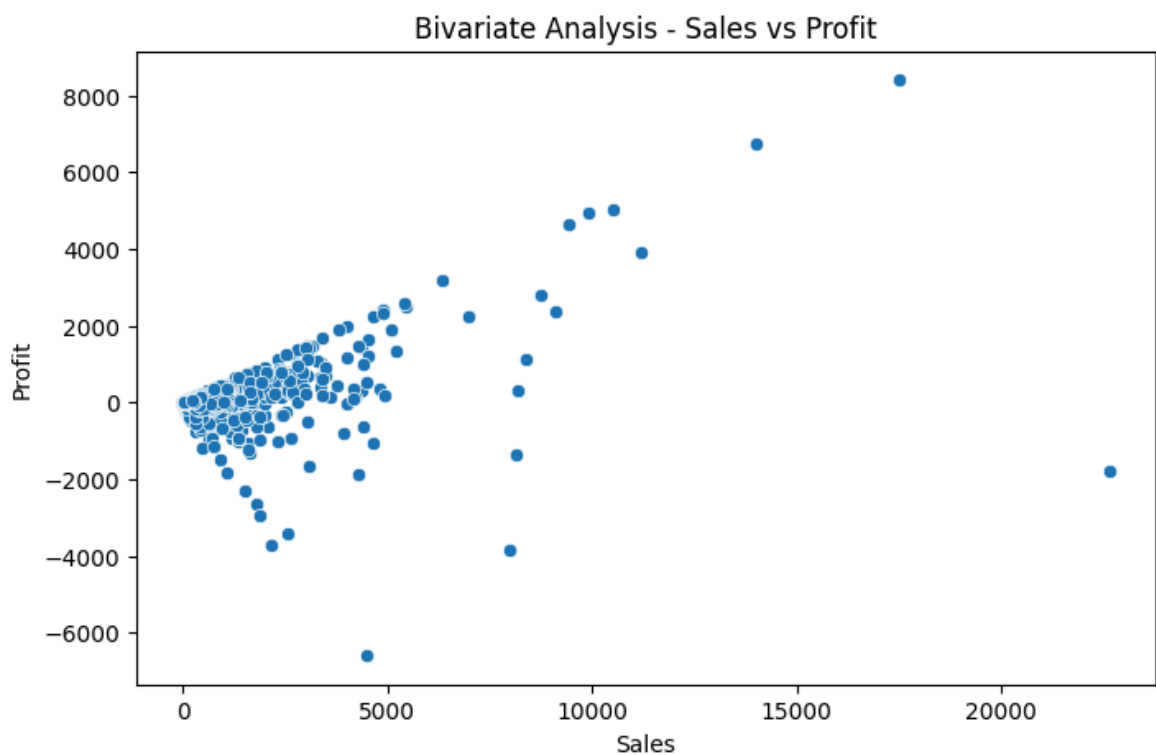
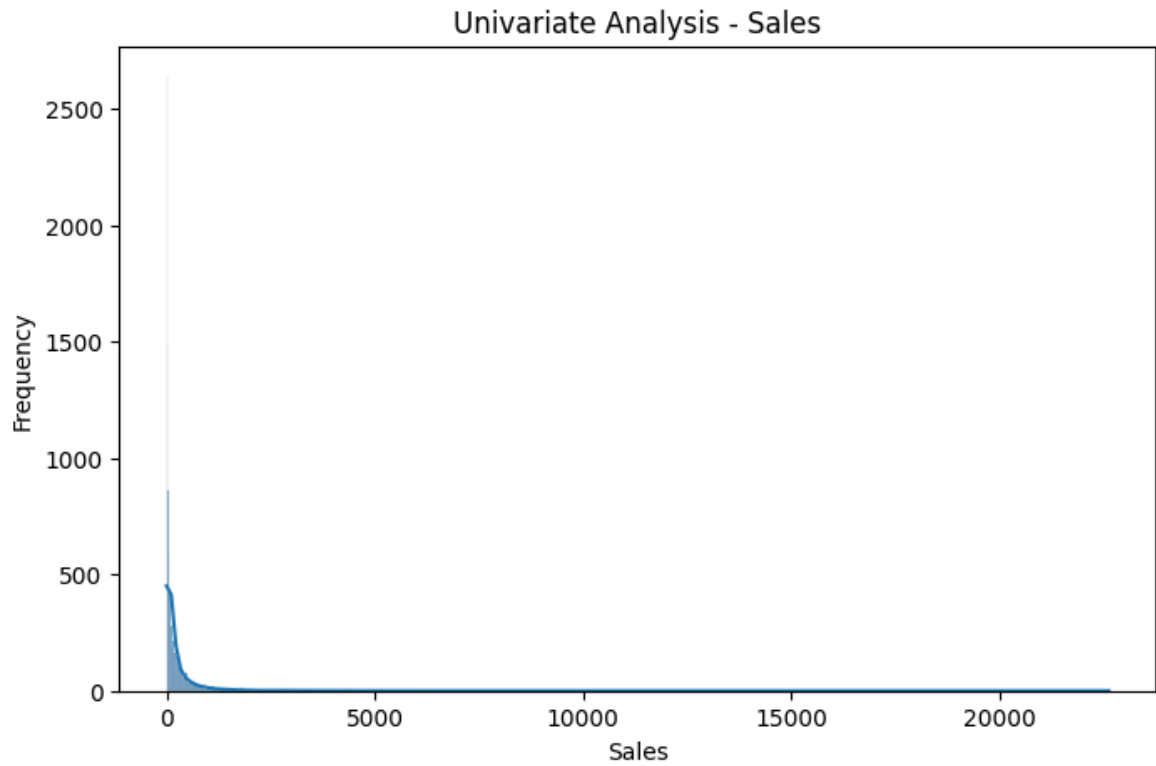
dtypes: float64(3), int64(3), object(15)

memory usage: 1.6+ MB

Statistical Summary:

	Row ID	Postal Code	Sales	Quantity	Discount \
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203
std	2885.163629	32063.693350	623.245101	2.225110	0.206452
min	1.000000	1040.000000	0.444000	1.000000	0.000000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000

	Profit
count	9994.000000
mean	28.656896
std	234.260108
min	-6599.978000
25%	1.728750
50%	8.666500
75%	29.364000
max	8399.976000

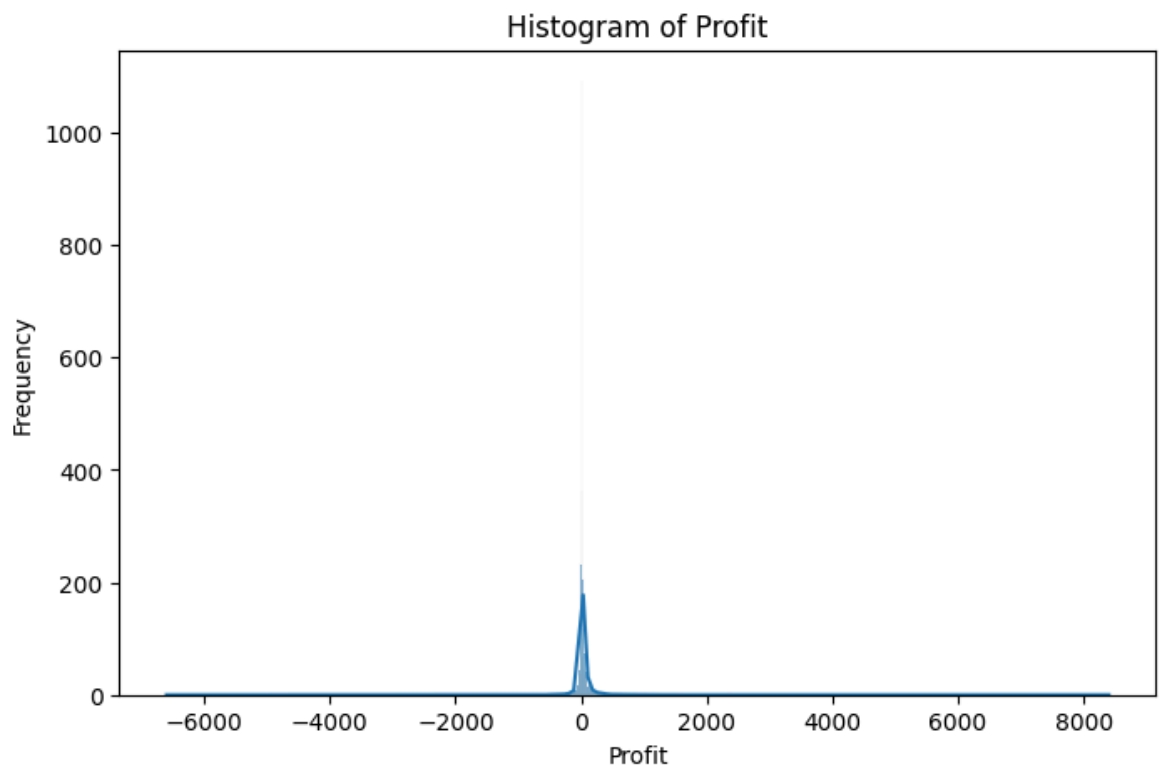
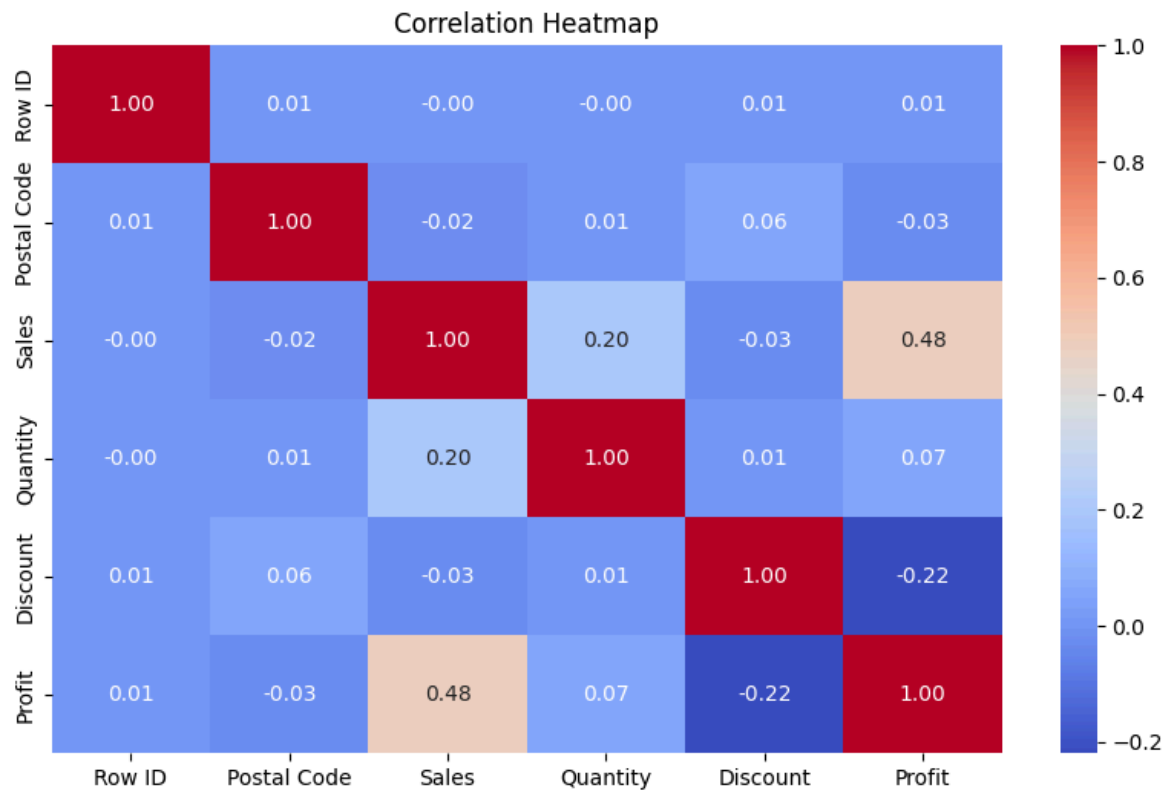


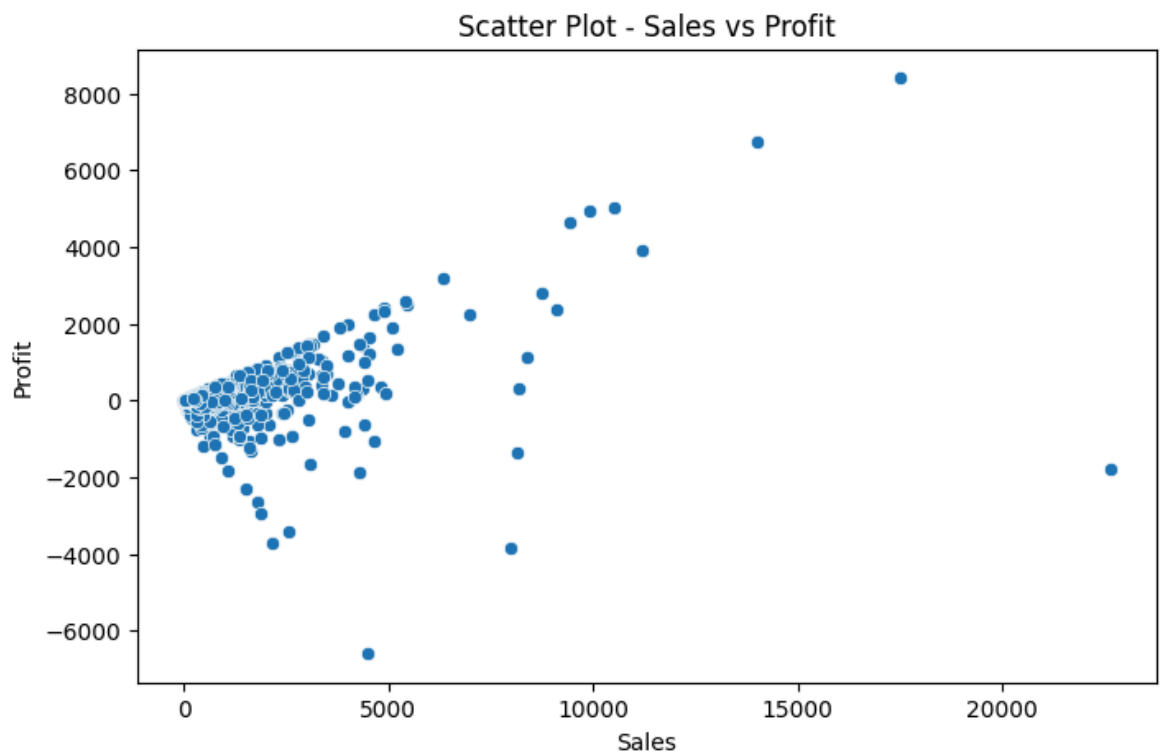
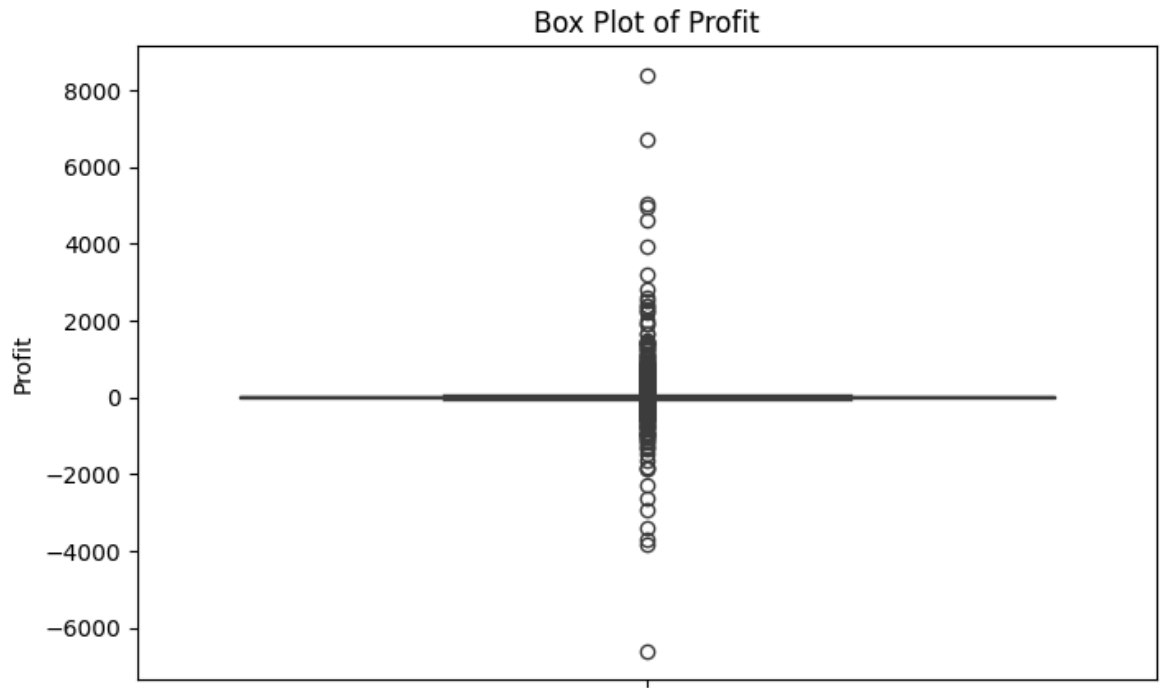
Numerical Columns:

['Row ID', 'Postal Code', 'Sales', 'Quantity', 'Discount', 'Profit']

Correlation Matrix:

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
Row ID	1.000000	0.009671	-0.001359	-0.004016	0.013480	0.012497
Postal Code	0.009671	1.000000	-0.023854	0.012761	0.058443	-0.029961
Sales	-0.001359	-0.023854	1.000000	0.200795	-0.028190	0.479064
Quantity	-0.004016	0.012761	0.200795	1.000000	0.008623	0.066253
Discount	0.013480	0.058443	-0.028190	0.008623	1.000000	-0.219487
Profit	0.012497	-0.029961	0.479064	0.066253	-0.219487	1.000000





Missing Values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

Duplicate Records:

0

Data Types:

Row ID	int64
Order ID	object
Order Date	object
Ship Date	object
Ship Mode	object
Customer ID	object
Customer Name	object
Segment	object
Country	object
City	object
State	object
Postal Code	int64
Region	object
Product ID	object
Category	object
Sub-Category	object
Product Name	object
Sales	float64
Quantity	int64
Discount	float64
Profit	float64

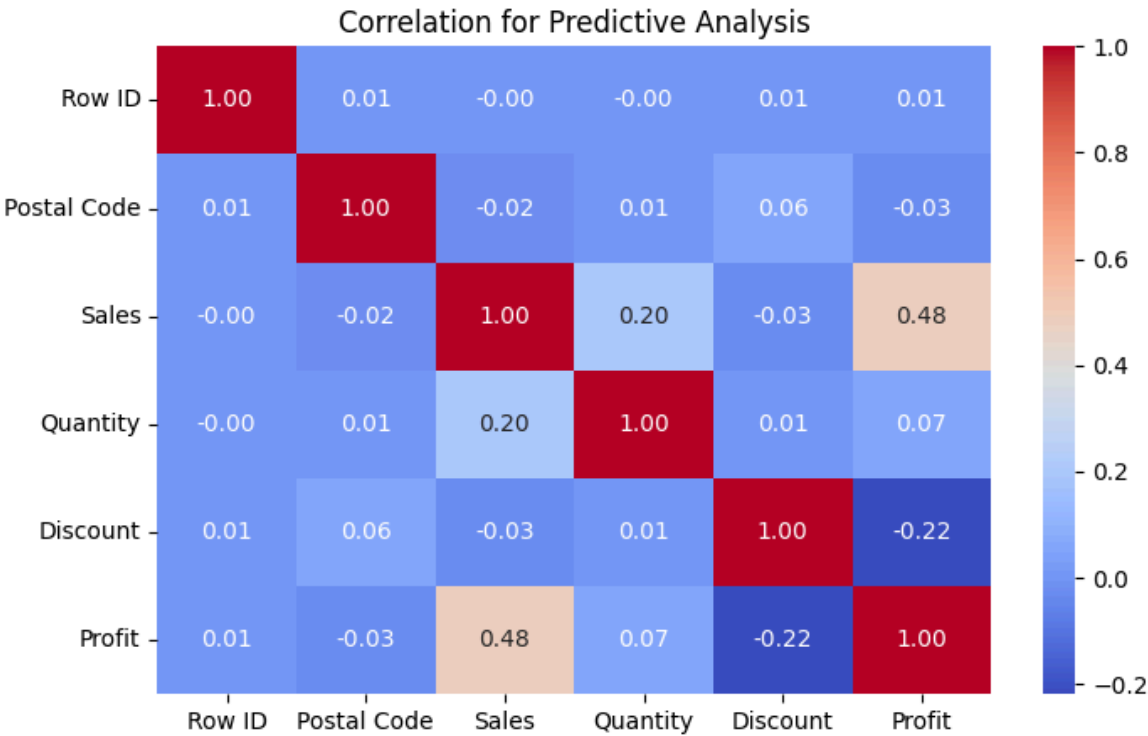
dtype: object

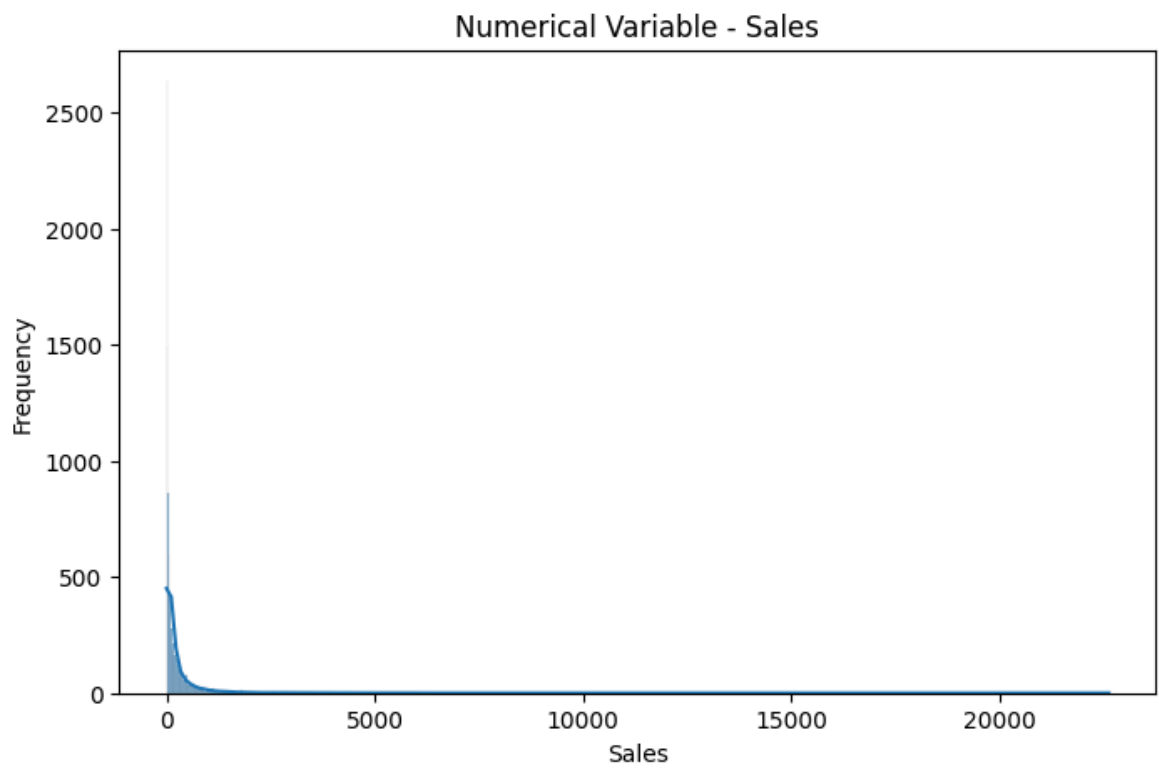
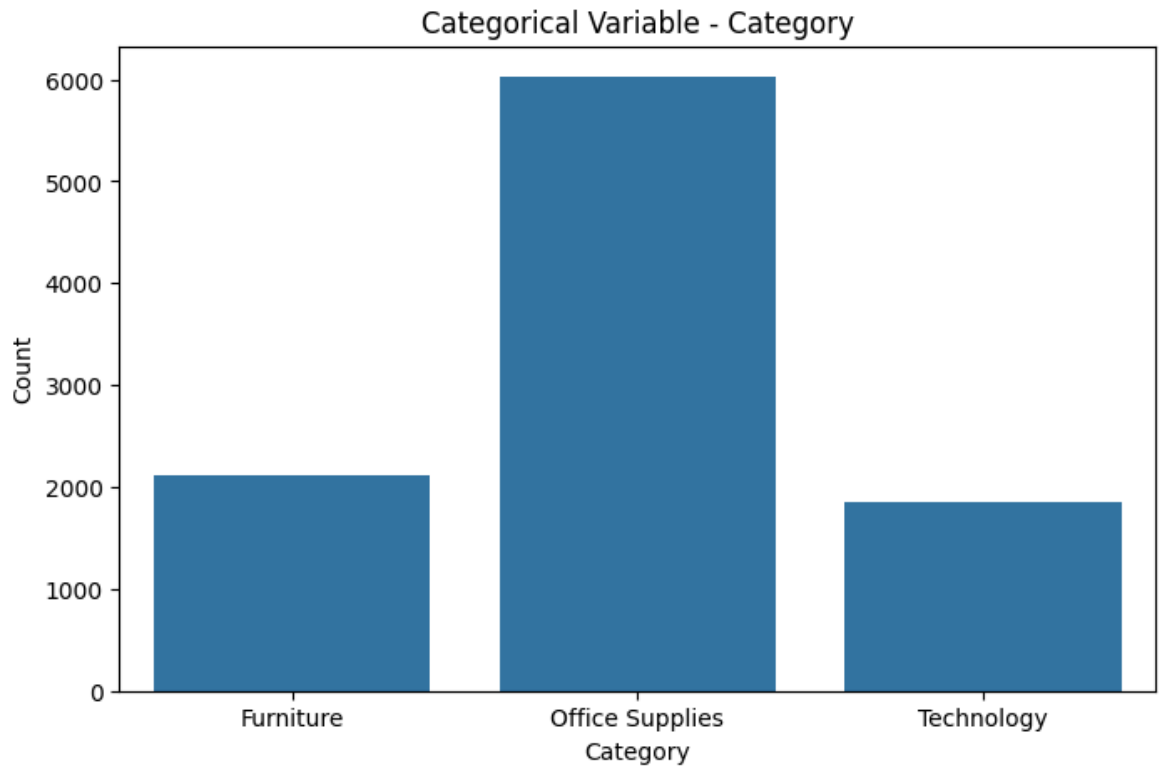
Unique Values:

Row ID	9994
Order ID	5009
Order Date	1237
Ship Date	1334
Ship Mode	4
Customer ID	793
Customer Name	793
Segment	3

```
Country      1
City         531
State        49
Postal Code  631
Region       4
Product ID   1862
Category     3
Sub-Category 17
Product Name 1850
Sales        5825
Quantity     14
Discount     12
Profit       7287
dtype: int64
```

```
Correlation with Profit:
Profit      1.000000
Sales       0.479064
Quantity    0.066253
Row ID      0.012497
Postal Code -0.029961
Discount    -0.219487
Name: Profit, dtype: float64
```





Sales by Category:

Category

Furniture	741999.7953
Office Supplies	719047.0320
Technology	836154.0330

Name: Sales, dtype: float64

Profit by Category:

Category

Furniture	18451.2728
Office Supplies	122490.8008
Technology	145454.9481

Name: Profit, dtype: float64

Sales by Region:

Region

Central	501239.8908
East	678781.2400
South	391721.9050
West	725457.8245

Name: Sales, dtype: float64

Profit by Region:

Region

Central	39706.3625
East	91522.7800
South	46749.4303
West	108418.4489

Name: Profit, dtype: float64

Sales by Segment:

Segment

Consumer	1.161401e+06
Corporate	7.061464e+05
Home Office	4.296531e+05

Name: Sales, dtype: float64

Profit by Segment:

Segment

Consumer	134119.2092
Corporate	91979.1340
Home Office	60298.6785

Name: Profit, dtype: float64

Highest Sales Category:

Technology

Highest Profit Category:

Technology

Highest Sales Region:

West

Highest Profit Region:

West

Feature Correlation with Profit:

Profit	1.000000
Sales	0.479064
Discount	0.219487

```
Quantity      0.066253
Postal Code    0.029961
Row ID        0.012497
Name: Profit, dtype: float64
```

Features with correlation greater than 0.3:

```
Profit      1.000000
Sales       0.479064
Name: Profit, dtype: float64
```

```
Number of Rows: 9994
Number of Columns: 21
```

```
Memory Usage:
9.231496810913086 MB
```

```
Sample Dataset Shape:
(1000, 21)
```

EDA completed successfully!